

Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

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Contents

1	Load Data	2
2	Preprocessing	2
3	Descriptive Checks	2
4	Primary: Per-Story Asymmetry Magnitude	3
5	Supplemental: Within-Condition Directional Asymmetry	4
6	Supplemental: Cross-Condition Directional Model	5
7	Supplemental: Directional Scatter Visualizations	7
8	Summary	7

This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing together

Important structural note:

- **author_1** / **author_2** are now treated as **slot 1** / **slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (**analysis_turn** 1 to 9).

Primary cross-condition object. Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$$

where $r_{1 \rightarrow 2, i}$ is the per-story Pearson correlation between same-turn slot 1 and slot 2 valence, and $r_{2 \rightarrow 1, i}$ is the correlation between slot 2's previous-turn valence and slot 1's current-turn valence. The main question is whether A_{valence} differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

1 Load Data

```
## Data availability:

## # A tibble: 6 x 4
##   file                                `human-ai` `human-human` `ai-ai`
##   <chr>                                <lgl>      <lgl>      <lgl>
## 1 dyadic_sentiment_scores.parquet      TRUE      TRUE      TRUE
## 2 interaction_level_surface_metrics.parquet TRUE      TRUE      TRUE
## 3 novelty_scores.csv                   TRUE      TRUE      TRUE
## 4 semantic_exploration_binned.parquet   TRUE      TRUE      TRUE
## 5 story_embeddings_interaction_level.parquet TRUE      TRUE      TRUE
## 6 interaction_level_stories_filtered.csv TRUE      TRUE      TRUE
```

Table 1: Loaded exchange-aligned valence data by condition

condition	n_stories	n_turns	min_turn	max_turn
Human-AI	100	854	1	9
Human-Human	31	279	1	9
AI-AI	39	351	1	9

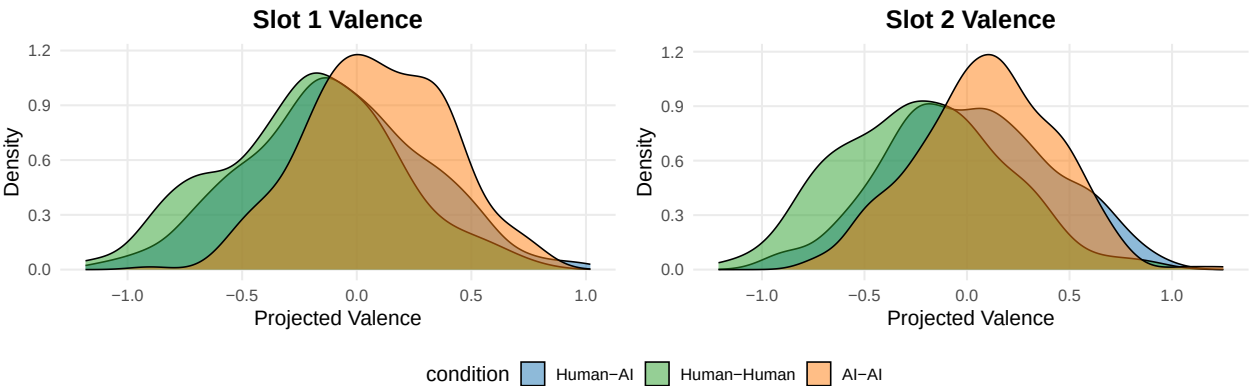
2 Preprocessing

```
## Analysis rows: 2628
## Stories: 170
```

3 Descriptive Checks

Table 2: Mean projected valence by condition

condition	n_stories	n_turns	slot_1_mean	slot_2_mean
Human-AI	100	754	-0.092	0.017
Human-Human	31	248	-0.214	-0.223
AI-AI	39	312	0.089	0.096



4 Primary: Per-Story Asymmetry Magnitude

Stories with both directional correlations available:

Table 3: Per-story A_valence sample sizes by condition

condition	n
Human-AI	100
Human-Human	31
AI-AI	39

Table 4: Per-story |Fisher-z asymmetry| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	100	0.551	0.466	0.436
Human-Human	31	0.457	0.400	0.408
AI-AI	39	0.421	0.348	0.371

One-way ANOVA: A_valence ~ condition

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2  0.561  0.2807   1.517  0.222
## Residuals 167 30.902  0.1850
```

##

Kruskal-Wallis (robust): A_valence ~ condition

##

Kruskal-Wallis rank sum test

##

data: A_valence by condition

Kruskal-Wallis chi-squared = 2.7938, df = 2, p-value = 0.2474

##

Pairwise Wilcoxon (BH-adjusted):

##

Pairwise comparisons using Wilcoxon rank sum test with continuity correction

##

data: df_asym\$A_valence and df_asym\$condition

##

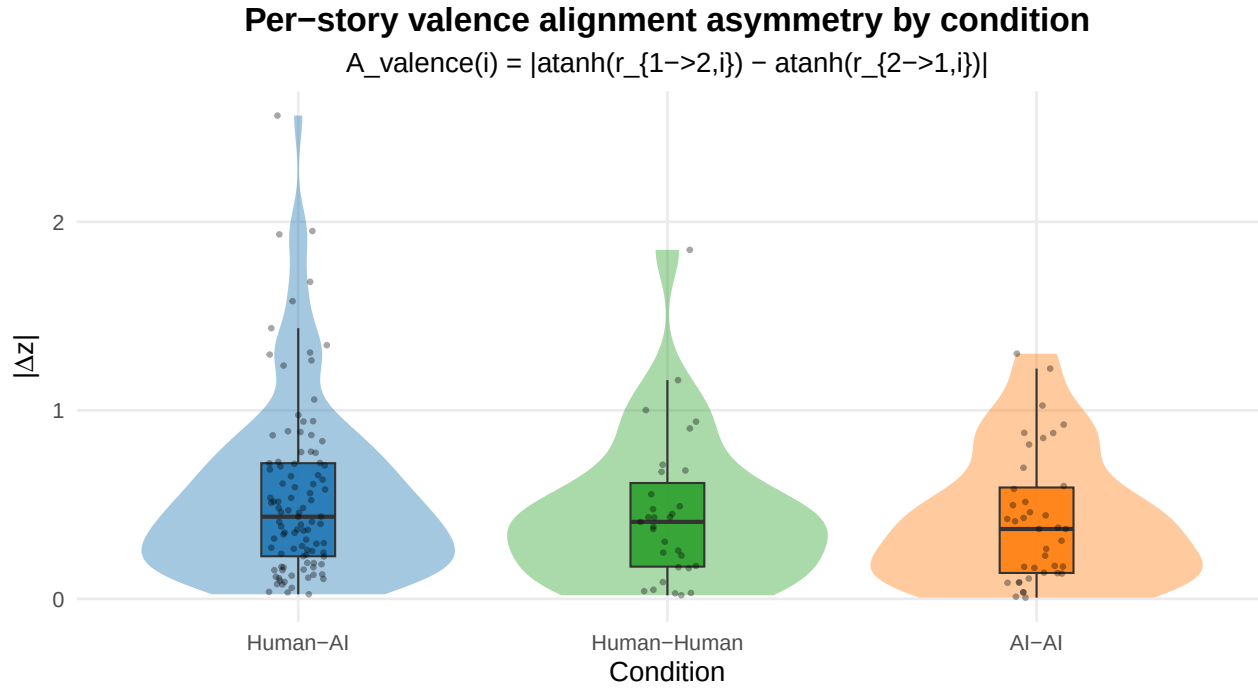
```
##           Human-AI Human-Human
```

```
## Human-Human 0.45      -
```

```
## AI-AI       0.38      0.73
```

##

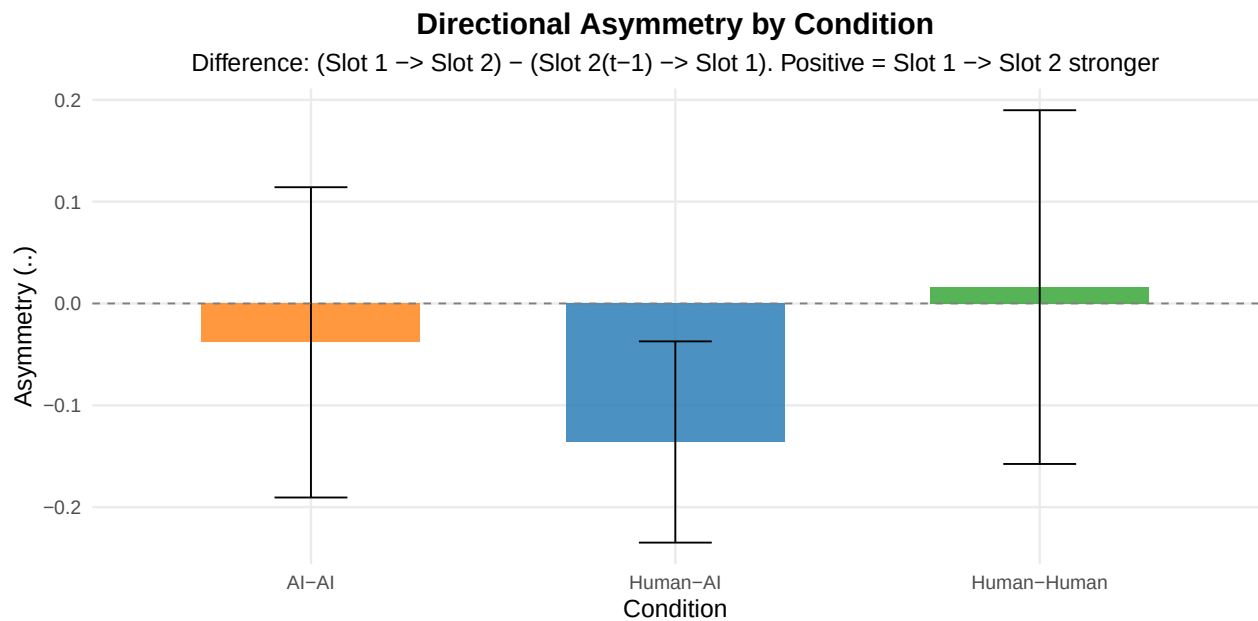
P value adjustment method: BH



5 Supplemental: Within-Condition Directional Asymmetry

Table 5: Within-condition alignment asymmetry by slot direction

condition	slope_slot1_to_slot2	slope_slot2_to_slot1	difference	se	t	p	sig
Human-Human	0.143	0.159	0.016	0.089	0.181	0.856	
Human-AI	0.216	0.080	-0.136	0.050	-2.698	0.007	**
AI-AI	0.193	0.154	-0.038	0.078	-0.491	0.623	



6 Supplemental: Cross-Condition Directional Model

```
## Full model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction * condition +
##          (1 + predictor_z | conversation_id)
## Data: df_std
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 7083.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.63406 -0.73863 -0.00121  0.72790  2.55633
##
## Random effects:
## Groups           Name          Variance Std.Dev. Corr
## conversation_id (Intercept) 0.00000  0.0000
##                  predictor_z 0.02047  0.1431   NaN
## Residual                0.83358  0.9130
## Number of obs: 2628, groups: conversation_id, 170
##
## Fixed effects:
##
##              Estimate Std. Error    df
## (Intercept)    1.558e-19  1.991e-02 2.452e+03
## predictor_z     1.578e-01  2.467e-02 1.627e+02
## alignment_direction1 1.316e-18  1.991e-02 2.452e+03
## condition1     -4.447e-18  2.410e-02 2.452e+03
## condition2     -1.147e-17  3.093e-02 2.452e+03
## predictor_z:alignment_direction1 2.635e-02  2.130e-02 2.452e+03
## predictor_z:condition1 -9.380e-03  2.982e-02 1.669e+02
## predictor_z:condition2 -6.411e-03  3.832e-02 1.613e+02
## alignment_direction1:condition1 2.141e-18  2.410e-02 2.452e+03
## alignment_direction1:condition2 1.228e-17  3.093e-02 2.452e+03
## predictor_z:alignment_direction1:condition1 4.166e-02  2.581e-02 2.452e+03
## predictor_z:alignment_direction1:condition2 -3.440e-02  3.308e-02 2.452e+03
##
##              t value Pr(>|t|)
## (Intercept)    0.000    1.000
## predictor_z     6.396 1.62e-09 ***
## alignment_direction1 0.000    1.000
## condition1      0.000    1.000
## condition2      0.000    1.000
## predictor_z:alignment_direction1 1.237    0.216
## predictor_z:condition1 -0.315    0.754
## predictor_z:condition2 -0.167    0.867
## alignment_direction1:condition1 0.000    1.000
## alignment_direction1:condition2 0.000    1.000
## predictor_z:alignment_direction1:condition1 1.614    0.107
## predictor_z:alignment_direction1:condition2 -1.040    0.299
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```

##          (Intr) prdct_ algn_1 cndtn1 cndtn2 pr_:_1 prd_:1 prd_:2 al_1:1
## predictor_z  0.000
## algnmnt_dr1  0.000  0.000
## condition1  -0.442  0.000  0.000
## condition2   0.266  0.000  0.000 -0.467
## prdctr_z:_1  0.000  0.000  0.000  0.000  0.000
## prdctr_z:c1  0.000 -0.445  0.000  0.000  0.000  0.000
## prdctr_z:c2  0.000  0.267  0.000  0.000  0.000  0.000 -0.466
## algnmnt_1:1  0.000  0.000 -0.442  0.000  0.000  0.000  0.000  0.000
## algnmnt_1:2  0.000  0.000  0.266  0.000  0.000  0.000  0.000  0.000 -0.467
## prdct_:_1:1  0.000  0.000  0.000  0.000  0.000 -0.439  0.000  0.000  0.000
## prdct_:_1:2  0.000  0.000  0.000  0.000  0.000  0.264  0.000  0.000  0.000
##          al_1:2 p_:_1:1
## predictor_z
## algnmnt_dr1
## condition1
## condition2
## prdctr_z:_1
## prdctr_z:c1
## prdctr_z:c2
## algnmnt_1:1
## algnmnt_1:2
## prdct_:_1:1  0.000
## prdct_:_1:2  0.000 -0.467
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

##
## Alignment slopes by condition and direction:

##   condition   alignment_direction   predictor_z.trend      SE  df lower.CL
## Human-AI      Slot 1 → Slot 2           0.2164 0.0385 512  0.14081
## Human-Human    Slot 1 → Slot 2           0.1433 0.0671 463  0.01147
## AI-AI          Slot 1 → Slot 2           0.1927 0.0598 463  0.07510
## Human-AI      Slot 2(t-1) → Slot 1       0.0804 0.0385 512  0.00479
## Human-Human    Slot 2(t-1) → Slot 1       0.1594 0.0671 463  0.02755
## AI-AI          Slot 2(t-1) → Slot 1       0.1545 0.0598 463  0.03692
## upper.CL
##   0.292
##   0.275
##   0.310
##   0.156
##   0.291
##   0.272
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

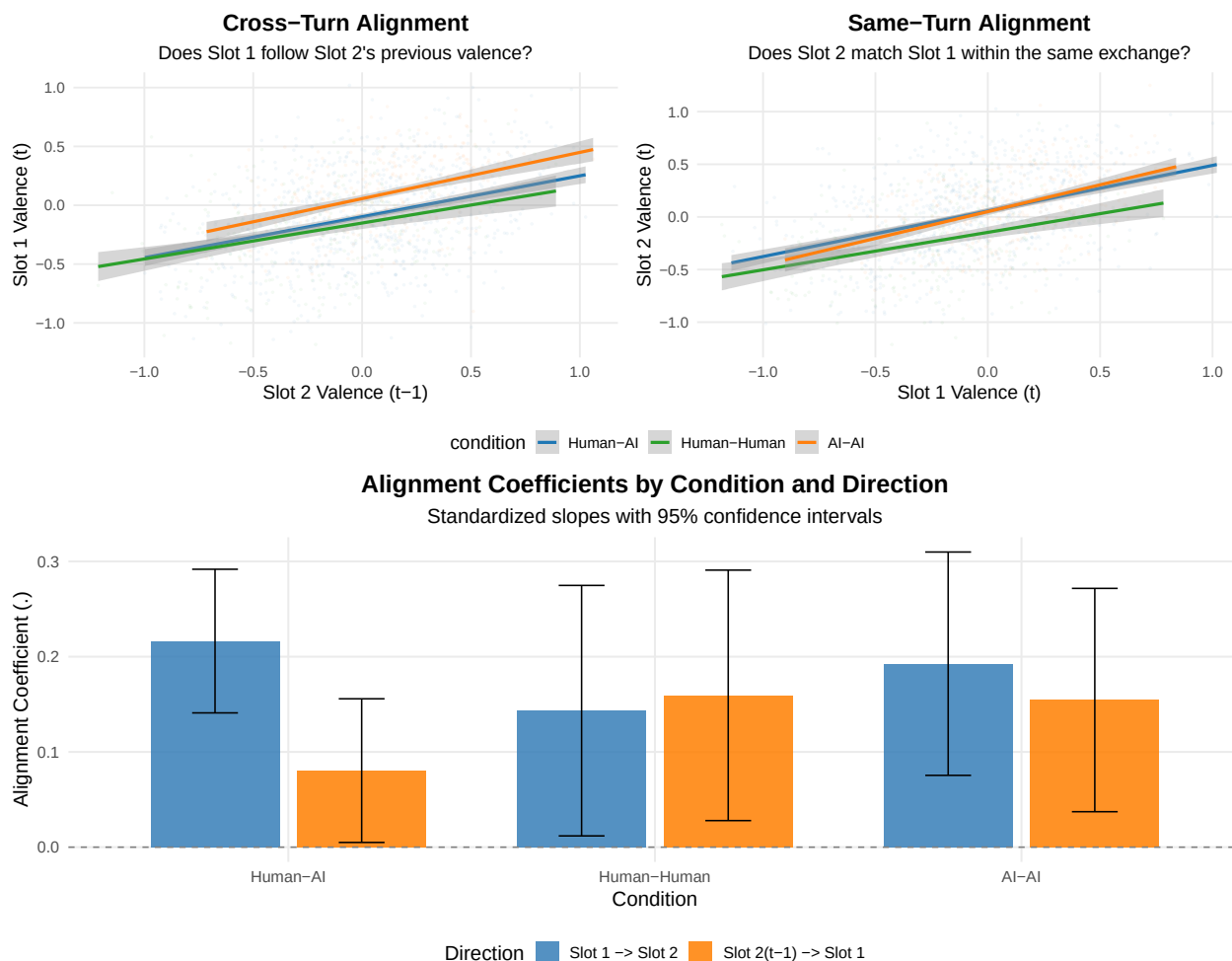
##
## Within-condition direction contrasts:

## condition = Human-AI:
##   contrast                                estimate      SE  df t.ratio p.value
## (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2 -0.1360 0.0505 2283  -2.694  0.0071
##
## condition = Human-Human:

```

```
## contrast                estimate      SE    df t.ratio p.value
## (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2  0.0161 0.0877 2283   0.183  0.8544
##
## condition = AI-AI:
## contrast                estimate      SE    df t.ratio p.value
## (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2 -0.0382 0.0781 2283  -0.489  0.6251
##
## Degrees-of-freedom method: kenward-roger
```

7 Supplemental: Directional Scatter Visualizations



8 Summary

Primary cross-condition result: per-story $A_{\text{valence}} = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Supplemental views retained:

- within-condition directional slope LMMs (two pairing rules)
- full cross-condition LMM with direction 00d7 condition interactions
- directional scatterplots

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)
- Human-AI decomposition (`speaker_type` \u00d7 `speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-26 15:53:53.032638